WORKING TITLE

by

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Abstract

My thesis is about bla bla

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Thanks!

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2. Main Contents
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   (b) Showing the model of infinity operads follows most of the axioms of an ∞-cosmos.

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Chapter 2

Background

2.1 Multicategories

EXPLAIN THE SOURCE OF THIS MATERIAL. "Multicategories were first defined in ..." "Our exposition follows ..."

Multicategories are a generalization of categories. In categories, arrows have a single source/domain to a single target/codoamin, in a multicategories the arrows have a finite sequence of sources/inputs to a single target/output.

Definition 2.1.1. A *multicategory* $\mathcal{M}$ consist of a set S of objects and for each $n \geq 0$ and each sequence $s_1, \ldots, s_n, t$ of objects a set

$$\mathcal{M}(s_1, \ldots, s_n; t)$$

of multiarrows/operations, thought of as taking n inputs of objects $s_1, \ldots, s_n$ respectively to the object output t. Moreover, there are three kinds of structure maps:

- For each object s, an identity/unit $1_s \in \mathcal{M}(s; s)$

- For any sequence of objects $s_1, \dots, s_n, t$ and any n-tuple of sequences $d^i_1, \dots, d^i_{k_i}$ of objects for $i = 1, \dots, n$, a composition function of multiarrows/operations

  $$\gamma : \mathcal{M}(s_1, \dots, s_n; t) \times \prod_{i=1}^{n} \mathcal{M}(d^i_1, \dots, d^i_{k_i}; s_i) \to \mathcal{M}(d^1_1, \dots, d^1_{k_1}, d^2_1, \dots, d^n_{k_n}; t),$$

  usually written $\gamma(p, q_1, \dots, q_n) = p \circ (q_1, \dots, q_n)$ or even just $p(q_1, \dots, q_n)$

Furthermore these structure maps have to satisfy a number of axioms, which express that composition is unital and associative. **ADD SPECIFIC CITATION WHERE THE FULL DEFINITION APPEARS.**

Remark 2.1.2. Let $\mathcal{M}$ be a multicategory, $f \in \mathcal{M}(s_1, \dots, s_n; t)$ and $g \in \mathcal{M}(d_1, \dots, d_l; s_i)$ for some $1 \leq i \leq n$. It will be convenient to have the following notation:

$$f \circ_i g := f(1_{s_1}, \dots, g, \dots, 1_{s_n})$$

Thus, $f \circ_i g$ is the composition of f with g in the ith variable.

Example 2.1.3.

(a) You can think of the multiarrows in $\mathcal{M}(s_1, \dots, s_n; t)$ as operations, taking inputs of types $s_1, \dots, s_n$ respectively to an output of type t. **FIX A SINGLE SET X? and then n-ary multiarrows are operations on X.** For example, define a multicategory $\mathcal{M}$ whose objects are sets and whose multiarrows are defined by setting $\mathcal{M}(X_1, \dots, X_n; Y)$ to be the set of functions

$$p : X_1 \times \cdots \times X_n \to Y$$

Composition in $\mathcal{M}$ is defined as composition of functions.

(b) If $\mathcal{C}$ is a symmetric monoidal category and C is a set of objects of $\mathcal{C}$, then one obtains a multicategory with this set of objects by defining $\mathcal{M}(c_1, \ldots, c_m; t)$ to be the set of morphisms $c_1 \otimes \cdots \otimes c_n \to t$ in $\mathcal{C}$. We will sometimes write $\mathcal{C}^{\otimes}$ for the multicategory obtained in this way, when C is the set of all objects of $\mathcal{C}$ (granted it is small).

(c) A (small) category $\mathcal{C}$ can be considered as a multicategory $\mathcal{C}_m$ which only have unary arrows. I.e. the set $\mathcal{C}_m(c_1, \ldots, c_n; c)$ is always empty unless $n = 1$.

We can observe that multicategories from a category $\mathcal{MC}$at.

Definition 2.1.4. For two multicategories $\mathcal{M}$ and $\mathcal{N}$, a *multifunctor* $\phi : \mathcal{M} \to \mathcal{N}$ consist of:

- A function $f : O_{\mathcal{M}} \to O_{\mathcal{N}}$ between the corresponding sets of objects
- For each sequence $s_1, \ldots, s_n, t$ of objects of $\mathcal{M}$, a map

$$\phi_{s_1,\ldots,s_n,t} : \mathcal{M}(s_1, \ldots, s_n; t) \to \mathcal{N}(f(s_1), \ldots, f(s_n); f(t)).$$

These maps being compatible with compositions and units in the obvious way.

2.1.1 Trees

Definition 2.1.5. A *tree* T consist of a finite set V of *vertices*, a nonempty finite set E of *edges*, a distinguished element $r \in E$ called the *root*, together with the following data:

(1) a function $I : E - \{r\} \to V$, which we think of as assigning to an edge e the vertex $I(e)$ of which is an input.

(2) A function $O : V \to E$, assigning to each vertex v its output edge $O(v)$

variable

References

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